

Physical Sciences Ethics Committee

CS Fast-Track Ethical Approval Form

How to Use This Form

You can make a copy of the form for editing through File > Make a copy. The copy then can be shared between the student and the advisor and collaboratively filled online. If you prefer to fill the form offline, the file can be downloaded through File > Download in a variety of formats including Microsoft Word (.docx) and OpenDocument Format (.odt). Finalised forms can then be exported as PDF through File > Download and submitted following the guidance on submission.

Notes before You Start

It is strongly recommended that you consult the guidance on the [CS Ethical Approval webpage](#). Non-compliance with the available guidance will lead to delays in the approval process.

This fast-track form is for **taught students** only. The form needs to be filled by the student and their advisor (or module leader if the project is a module assignment).

Research students and staff must complete the Full Ethical Approval Form found in the [CS Ethical Approval webpage](#).

Students: You need to discuss the ethical considerations of your project with your project advisor (or module leader if the project is a module assignment) and, if necessary, fill in a full ethics form to be submitted to the Physical Sciences Ethics Committee.

Advisors (or Module Leaders): You need to review and approve the completed form. Please ensure you are familiar with the University's [Code of practice and principles for good ethical governance](#) to guide your student effectively. Please seek guidance from the Departmental Ethics Officer(s) if you are uncertain about any ethical issue arising from this application.

Section 1. Potential Ethical Issues

Does your project involve any of the following? Please mark Yes or No for **all** issues.

No.	Issue	Yes	No
1.1	Human participants (adults or children)	✓	
1.2	Personal data (whether primary data collected through surveys/questionnaires or from online sources or secondary data from available datasets) – please consult the definition of personal data	✓	
1.3	Datasets that require explicit permission from the data provider		✓
1.4	Applications that could potentially involve unethical practice, including potential dual-use applications (e.g. projects involving tools or data that can be used for unethical purposes e.g. to attack systems)		✓
1.5	Funding sources or collaboration with potential to adversely affect existing relationships or bring the University or Department into disrepute (e.g. projects related to gambling, dark market, etc.)		✓
1.6	Restrictions on dissemination (e.g. not being allowed to publish certain datasets or results)		✓
1.7	Military or defence context		✓
1.8	Overseas countries under regimes with poor human rights record or identified as dangerous by the Foreign & Commonwealth Office		✓
1.9	Human material (e.g. tissue or fluid samples), vertebrates, especially mammals and birds, or any other organisms not previously mentioned		✓

If you answered **No** to all the above, you do not need ethical approval, so you do not need to continue filling in this form and you do not need to submit it for approval.

If you answered **Yes** to any of the above, you must complete this Fast-Track Ethical Approval Form, get it signed off by your project advisor (or module leader), and submit it for approval to the Departmental Ethics Officers.

The rest of this form is designed to verify specific conditions. If certain conditions are not satisfied, the form will guide you to complete a Full Ethical Approval Application, to be approved by the Physical Science Ethics Committee.

Section 2. Project Information

Student Name	Ella Daramola
Course Title	BENG in Computer Science
Project Advisor	John Oyekan
Project Title	Mitigating Implicit Bias in Clinical Decision Support Systems through Fatigue detection

Course Mode of Delivery:

☒	On Campus	☐	Online
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Project Type:

☒	Undergraduate Project	☐	Postgraduate Project
☐	Undergraduate Module Assignment	☐	Postgraduate Module Assignment
☐	Other - Please specify:		

Project Description

2.1. Provide a clear statement of your research questions or your experimental hypotheses.

My study proposes a way to reduce healthcare professionals from making biased medical decisions when they are tired or overwhelmed through the detection of fatigue.

This study aims to draw a conclusion that bias can be exacerbated by fatigue(through research papers) and therefore find out if mitigating fatigue through technology can reduce fatigue errors leading to bias.

2.2. Briefly explain the methodology of your study. Give sufficient detail that a non-expert in the subject can understand what you are proposing to do, especially with regards to your project data collection and analysis.

Overall study

Observing Digital movement: Instead of using intrusive cameras or brain-scanners, my system observes how a person moves their mouse. When people are tired, their movements become jittery, less direct, and their mouse rhythm changes. We call this mouse Telemetry. The baseline mouse telemetry is collected by asking participants to move their mouse for 30 seconds without any task given, thus establishing their normal patterns.

The AI Brain: This telemetry data is sent to a Machine Learning model(trained on the fatigue dataset). The model is trained to recognize if a doctor is fatigued and subsequently triggers the UI. As well as a similar bias test (similar to Project Implicit: Health Professional Dataset, which will be justified as a standard bias test) for accurate calculation of intervention.

The final task of this study where the results of the Machine Learning model can be extracted from is a simple categorization task(that can be completed by anyone). Participants are informed of the rules beforehand e.g if a patient's blood pressure is higher than 90, sort them into the "high risk" category(done by clicking a button). This simulates a mock clinical environment without using real patient data or real medical knowledge.

The System once detecting fatigue changes the colour of the bordering rim of the page (color shifts proven to bring attention back through research), this **Hightlights** and brings users focus back to the task at hand. All color changes designed to be non-abrasive or shocking to the user.

Participants

General Population (Healthy Adults): The study targets a balanced group of healthy, adults recruited from the university **ABOVE THE AGE OF 18**. The study aims to prove that the general population experiences around the same range of fatigue and bias so anyone is suitable for the study(through research papers) therefore exclusive need of doctors or medical people are not needed. The study aims to get 10 people.

The only exclusion would be the visually and physically impaired as they would need to be able to appropriately see the and use the interface(through the mouse).

More details on participants in a later section

Section 3. External Datasets

If you are not planning to use any external datasets in your project, please mark the following and skip to Section 4.

	I am <u>not</u> using any external datasets in my project.
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3.1. If you are using external datasets, please identify the dataset(s). Give sufficient information about each dataset, including relevant names, locations, and any links to the repositories.

External Dataset 1: Project Implicit: Health Professional Dataset

I will utilize the "Project Implicit: Health Professional Dataset" hosted by the Open Science Framework (OSF). This dataset contains raw results from hundreds of thousands of Implicit Association Tests (IAT) taken by healthcare providers. I will use this to establish a baseline of how to quantify bias

<https://osf.io/26pkd/overview>

External Dataset 2: Human Decision Fatigue Dataset (Kaggle)

I am utilizing the Human Decision Fatigue Behavioral Dataset available on Kaggle, uploaded by cognitive tasks over extended periods. Crucially, it tracks variables such as:

- Hours Awake: To simulate the onset of fatigue.
- Task Performance Accuracy: To establish a "Ground Truth" for fatigue.
- Reaction Times / Decision Speed

SONAL SHINDE. This dataset contains records of participants performing(b) License and Compliance

<https://www.kaggle.com/datasets/sonalshinde123/human-decision-fatigue-behavioral-dataset>

3.2. For each of the external datasets, please identify the licence or terms of use (or equivalent), provide a copy of the relevant clauses here, and justify why your project would be compliant with those terms. Provide sufficient information so that reviewers do not need to refer to the full terms.

Dataset 1

- License: This dataset is released under the Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) license, also known as CC0: Public Domain or CC BY 4.0 (Attribution).
- Compliance Justification: My project is strictly academic and non-commercial, directly aligning with the "Non-Commercial" clause. I will provide full attribution to the original researchers in all publications. Furthermore, the data is utilized in an aggregated, de-identified format, ensuring that no individual clinician can be identified, which complies with the UK GDPR and Data Protection Act 2018.

Dataset 2

- License: This dataset is released under the Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) license, also known as CC0: Public Domain or CC BY 4.0 (Attribution).
- Justification: Under these terms, the data is free to be used for academic and research purposes. My project complies by providing full attribution to the original author in the bibliography and using the data solely for the training and validation of my "Bias-Aware" ML model. As a non-commercial student project, this usage falls squarely within the "Fair Use" or "Educational Use" provisions of the license.

Both dataset licence: <https://creativecommons.org/publicdomain/zero/1.0/>

3.3. If the dataset includes data about or collected from humans, please provide evidence that the data was initially collected with consent.

Dataset 1

The Project Implicit data is collected via an online platform where every participant must read and sign an Informed Consent Form approved by the Harvard University Institutional Review Board (IRB). The consent form explicitly states that the de-identified data will be made available to the global scientific community for the purpose of

understanding implicit bias and improving healthcare outcomes. Evidence of this consent protocol is documented in the dataset's public "Data Dictionary" and "Ethics Statement" on the OSF platform. <https://osf.io/26pkd/overview>

Dataset 2

- The dataset is identified as synthetically generated for the purpose of modeling cognitive fatigue trends. As no actual human participants were involved in the creation of this specific CSV, it does not require individual consent forms, though it is modeled after validated human behavioral patterns."

Section 4. Informed Consent

If you do not have participants in your study, please mark the following and skip to Section 5.

	There are no participants in my study.
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If you have participants in your study, complete the following table. You need to submit your **participant information sheet** and **consent form**, as well as the draft of any **questionnaires** or **interview questions** along with this form.

If you answer **No** to any of the following, you must submit a Full Ethical Approval Form.

No.	Question	Yes	No
4.1	Will you inform the participants of the purpose of the study, the investigators, and any funding source?	✓	
4.2	Will you describe the procedures and the data collected to participants in advance, so that they are informed about what to expect? Note: Any personal or sensitive information, and any audio or video recording must be explained explicitly and in detail.	✓	
4.3	Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs? Note: If there is a plan to publish data, e.g. direct quotations, on an individual basis (rather than in aggregate), the exact anonymisation method needs to be explained clearly.	✓	
4.4	Will you inform participants of any possible risks of participation?	✓	

4.5	Will you inform participants that their participation is voluntary, that they may withdraw from the research at any time (even after the data collection, for a reasonable and specified time period) and for any reason without any negative consequences, and how they can request withdrawal?	✓	
4.6	Will you obtain and record explicit consent for participation?	✓	
4.7	Answer this question only if your research is observational, i.e. you plan to physically observe participants. Will you obtain and record explicit consent to being observed?		
4.8	Answer this question only if your research involves any audio or video recording. Will you obtain and record extra explicit consent to being audio or video recorded?		
4.9	Answer this question only if there is a plan to publish data, e.g. direct quotations, on an individual basis. Will you obtain and record extra explicit consent to any such publication clearly described to the participants?		
4.10	Answer this question only if you plan to contact the participants in the future. Will you obtain and record extra explicit consent to being contacted with the contact method specified for the participants?		

If you answered **Yes** to any of the above, this must be explicit in your supporting documents, e.g. consent forms, information sheets, and questionnaires.

Section 5. Protocol Issues

If you answer **Yes** to any of the following, you must submit a Full Ethical Approval Form.

No.	Question	Yes	No
5.1	Is your study designed to be challenging, disturbing, or invasive (physically or psychologically) for anyone including yourself?		✓
5.2	Will you deliberately mislead your participants?		✓
5.3	Could the methodologies or findings of your study damage the reputation of the University of York?		✓

5.4. If there are any obvious or inevitable adaptations of your research findings to ethically questionable aims, e.g. tools or data that can be used to attack systems, please explain the measures you are putting in place to limit such adaptations.

N/A

Section 6. Health and Safety

Identify any risks to the participants and state any precautions you will take to ensure their physical and mental health and safety. If you believe there are no risks, please state why.

There are no risks associated with this study. As this study doesn't require any physical activity to be done (study done seated with a computer) and is not deemed mentally strenuous (simple tasks not designed to induce fatigue).

Precautions and Safety Measures

- Voluntary Participation: All participants must sign an informed consent form before starting, and they are told exactly what the experiment involves.
- Stopping the Test: The study ends immediately if a participant states they want to stop.
- Data Privacy: All behavioral data and personal scores are kept safe and private to ensure the participants' digital security and.

Section 7. Vulnerable Groups

If you answer **Yes** to the following question, you must submit a Full Ethical Approval Form.

If your participants are **patients**, in addition to the Full Ethical Application you must follow the Guidelines for Ethical Approval of NHS Projects.

If your project involves any vulnerable group, you may also need to obtain satisfactory Disclosure and Barring Service (DBS) clearance (or equivalent for overseas students).

No.	Question	Yes	No
7.1	Does your project directly involve (e.g. as participants) any vulnerable		✓

	groups, e.g. children under 18, people with learning difficulties, people who have physical or mental health issues, patients (e.g. NHS patients)?		
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7.2. If any vulnerable groups are indirectly involved, please specify the group(s) and their involvement in detail below. Examples of indirect involvement include collecting data from parents about their children, or working with external datasets about vulnerable groups.

N/A

Section 8. Data Protection

If you answer **No** to any of the following, you must submit a Full Ethical Approval Form.

No.	Question	Yes	No	N/A
8.1	Any personal or sensitive data will be stored in password protected folders on University shared drives or computers or those of an approved partner.	✓		
8.2	Any hard copies of personal data (e.g. any hard-copy consent forms) will be stored in a secure place within University premises, or those of an approved partner.	✓		
8.3	Only the student and advisor will have access to any personal data (see definition) involved in the study. Note: Anonymised data may be shared to a wider group (typically for research purposes), but the consent form needs to make this clear.	✓		
8.4	The data will be preserved beyond the study in line with University policy and will be placed in the custody of the advisor at the end of the project, or destroyed in accordance with the retention policy provided to participants.	✓		
8.5	All data will be anonymised prior to analysis. Please state your method of anonymisation: Numerical Coding: Each participant will be assigned a unique number (e.g., Participant 001). This ID will be used for all IAT scores, mouse telemetry (τ), and triage results. Also the aggregate result will be published	✓		

Section 9. Personal Data

If your project involves working with personal data (see [definition](#)), briefly specify each of the following.

This applies both if (a) you have participants in your project (i.e. you are collecting primary personal data from humans) or (b) you are working with a dataset that includes personal data about humans (i.e. you are working with secondary data about humans).

9.1. Who are the target participants or data subjects? And how are you recruiting them or how were the data about them collected?

Target Participants

- General Population (Healthy Adults): The study targets a balanced group of healthy, adults recruited from the university(University of york) **ABOVE THE AGE OF 18**. The study aims to prove that the general population experiences around the same range of fatigue and bias so anyone is suitable for the study(through research papers) therefore exclusive need of doctors or medical people are not needed. The study aims to get 10 people.
- The only exclusion would be the visually and physically impaired as they would need to be able to appropriately see the and use the interface(through the mouse).
- University of York student would fall into this criteria(general population) as they would have the same levels of bias and fatigue as the average person and will explicitly be checked if they are above 18.

Recruitment and Data Collection Method

- Recruitment Strategy: Participants are recruited through university channels like through social media posts on society channels or word of the mouth.
- Compensation: Participants will not be compensated or rewarded, but will only do this study voluntarily if they want to contribute to the study.
- Informed Consent: All subjects are fully informed of the experimental details and must sign a consent form before any data collection begins.
- Primary Data Collection: Data is collected on campus in any suitable room (containing a desk and chair) through my own coded study.

9.2. Exactly what pieces of personal data (including demographic data) you are collecting or using from an existing dataset? Explain why you need each.

From individuals in the study: No personal data is collected as Fatigue measured by mouse

tortusity and bias result does not fall under personal data

From Fatigue dataset: no personal data is collected as variables like Hours Awake, Reaction Times / Decision Speed does not fall under personal data. Data in this study is considered synthetic

From Implicit bias dataset: No personal data containing racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union-membership, health, sex life and sexual orientation, genetics, Biometrics of participants is used. However participants opinions on age, gender and race was used to make a general bias score(chance or likelihood of bias)

9.3. How are you planning to publish your data and analysis results: on an aggregate basis or on an individual basis? If on an individual basis, explain why it is **necessary** to do so and if they will be anonymised.

All results from this study will be published on an aggregate basis. This means the data will be reported as group totals, averages, and general trends rather than identifying specific individuals. The data will be held for two weeks or until my dissertation is marked and will only be accessed by me (the investigator) and no one else. After which all data will be permanently deleted.

9.4. Is there a risk that the data or analysis results you publish can be de-anonymised? If yes, explain how you are planning to mitigate it.

There is a very low risk that data could be de-anonymized, but the study includes specific plans to prevent this.

Aggregated Reporting: We will only publish group trends, averages, and overall patterns rather than sharing individual raw mouse movement data or personal scores.

Identifier Removal: Names and personal details are removed and replaced with random identification numbers before any analysis or publication takes place.

9.5. If you have questionnaires or interviews, will you give participants the option of skipping any questions they do not want to answer? If any of your questions are mandatory, explain why they must be.

This study is not in the format of questionnaire or interviews. Participants will either consent or not consent to the study. Participants are also informed about the experiment details and must provide their signed consent before starting.

Section 10. Student Declaration

Please mark boxes below that apply and sign.

☒	I have considered the ethical implications of this project and have identified no significant ethical implications requiring a full ethical application submission to the Physical Sciences Ethics Committee.
☒	I will include all supporting documents (e.g. information sheet and consent form, drafts of questionnaires or interview questions) with this application.
☒	All the components specified in Section 4 are included explicitly in the information sheet / consent form submitted with this application.

Student Name	Emmanuella Daramola
Student Signature	Daramola
Date	20/04/2026

Section 11. Advisor (or Module leader) Approval

If this project is related to any previously approved work, please include the application reference for that work here and briefly explain the similarities and differences. This usually helps expedite the consideration by the ethics committee.

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Please mark boxes below that apply and sign.

☒	The student has taken all reasonable steps to ensure ethical practice in this study and I can identify no significant ethical implications requiring a full ethical application submission to the Physical Sciences Ethics Committee.
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✓	I have checked and approved all supporting documents required for this application.
✓	I understand that completion of this form indicates that from the ethical point of view I am willing to share responsibility for the work being conducted.

Advisor Name	John Oyekan
Advisor Signature	John Oyekan
Date	20/04/2026

Submission

Please make sure this form is completed in full, signed by both the student and the advisor, and accompanied by required supporting documents (e.g. participant information sheet and consent forms). Otherwise, the approval process is delayed.

Please convert this form to PDF after it is finalised and before submission.

Please submit this form in **PDF** and all supporting documents, preferably all in **PDF** (unless the file cannot be converted or printed to PDF), to the Departmental Ethics Officers through the submission form listed in the [CS Ethical Approval webpage](#).